

ENERGY ENGINEERING REVIEW BOARD

Product Requirements Document

Document Version	1.0 – Final for Hackathon
Date	September 2026
Status	MVP Release Candidate

EXECUTIVE SUMMARY

Energy Engineering Review Board (EERB) is an AI-powered preliminary engineering review system designed to identify technical inconsistencies in proposed renewable energy projects before detailed design and implementation begin.

Problem: Energy project teams frequently discover critical design flaws after expensive engineering work has commenced, resulting in significant cost overruns and timeline delays.

Solution: EERB provides automated preliminary technical review using a multi-agent AI system that independently analyzes project specifications, identifies conflicts, and produces traceable engineering assessments.

Impact: Reduces review time from 1-2 weeks to 2-5 minutes and costs from \$2,000-\$5,000 to \$10-\$50 per review.

Key Differentiators

- ✓ **Deterministic Math:** All calculations are verifiable Python code, not LLM hallucinations
- ✓ **Conflict Detection:** Automatically identifies disagreements between analysis agents
- ✓ **6-Agent Analysis:** Parallel evaluation from Load, PV, BESS, Spec, Critic, and Lead Engineer
- ✓ **Traceable Reports:** Complete audit trail with evidence sources and recommendations
- ✓ **Rapid Assessment:** 2-5 minutes vs. 1-2 weeks for manual review

THE PROBLEM

Energy projects currently discover critical design flaws AFTER expensive engineering work has commenced, resulting in massive cost overruns and delays.

Business Impact:

Metric	Traditional	With EERB	Improvement
Review Time	1-2 weeks	2-5 min	100-500x faster
Cost per Review	\$2,000-\$5,000	\$10-\$50	90% savings
Consistency	Variable	Systematic	Highly reliable

THE SOLUTION: EERB

EERB uses 6 specialized AI agents that work in parallel to analyze projects from different engineering perspectives:

Agent	Responsibility
Load Analyst	Analyzes electricity consumption patterns
PV Engineer	Evaluates solar system adequacy
BESS Engineer	Assesses battery storage capability
Spec Engineer	Validates technical documents
Critic	Identifies conflicts and assumptions
Lead Engineer	Synthesizes findings and recommendations

DEMONSTRATION: REAL DESIGN CONFLICT

Project: 500 kW Solar PV + 1 MWh Battery for Commercial Building

Objective: Reduce peak demand by 30%

THE CONFLICT EERB DETECTS:

Battery Duration (Deterministic Math):

Duration = Energy ÷ Power = 1,000 kWh ÷ 500 kW = **2.0 hours**

Peak Period Duration (from Load Analysis): 4.5 hours

Gap: 4.5 – 2.0 = **2.5 hours of unmet demand**

CONCLUSION: Battery CANNOT support full peak period. Design requires rework.

TECHNICAL ARCHITECTURE

Frontend: Streamlit + Plotly (interactive charts)

Backend: Python 3.10+, Pandas, NumPy

LLM: OpenAI GPT-3.5-turbo

Deployment: Streamlit Community Cloud

Version Control: GitHub

MARKET OPPORTUNITY

Target Market: 500+ Renewable Energy EPCs in US, 1000+ globally

Addressable Market: \$5M+ annually

MVP SUCCESS CRITERIA

- ✓ Application runs end-to-end without errors
- ✓ Demo project executes in <60 seconds
- ✓ Conflict detection accurately identifies design issue
- ✓ Reports download in readable format
- ✓ All calculations verifiable and auditable
- ✓ Code clean, documented, deployment-ready

CONCLUSION

EERB addresses a real, sizeable market need: rapid, affordable preliminary engineering review for renewable energy projects. The MVP demonstrates core value through conflict detection, deterministic calculations, and traceable analysis.

The MVP demonstrates core value through:

- Conflict detection (core differentiator)
- Parallel agent analysis
- Deterministic, verifiable math
- Traceable, auditable reports
- Rapid assessment (2-5 minutes)
- Actionable recommendations

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